

5.3 Solving the Matrix Inverse by Hand



Core skills you'll master in this section:

- ▶ The precondition for a square matrix to be invertible is that its determinant is nonzero.
- ▶ Delete the corresponding row and column to construct the minor matrix for each element.
- ▶ Taking the determinant of each minor matrix gives the minor.
- ▶ Adjust the minor's sign according to its position to obtain the cofactor.
- ▶ The cofactors form the cofactor matrix.
- ▶ Transposing the cofactor matrix gives the adjugate matrix.
- ▶ The inverse matrix can be obtained by dividing the adjugate matrix by the determinant.
- ▶ Review: computing the determinant via Laplace expansion.

For square matrices that are small (or have a special structure), we can solve for the inverse by hand. There are two commonly used methods: the adjugate matrix method and the elementary row operations method. This section introduces the adjugate matrix method; the elementary row operations method will be introduced near the end of this volume.

Readers who don't need to know how to solve a matrix inverse by hand can skip this section.

In this section, we'll start with the simple 2×2 matrix. We'll first review the basic concept of the matrix inverse, and explore its relationship with the determinant and the adjugate matrix. Then we'll gradually extend to the 3×3 matrix, explaining in detail how to find the inverse via the adjugate matrix, with a concrete worked example.

The example used in this section comes from the previous chapter's section on solving determinants by hand; so this section can also help us review the minor, the cofactor, and computing the determinant via Laplace expansion.

The Inverse of a 2×2 Matrix

Given the following 2×2 matrix A

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

A 's inverse matrix is

$$\mathbf{A}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Earlier in this chapter, we mentioned that the necessary and sufficient condition for the square matrix $\mathbf{A}$ to be invertible is that $\mathbf{A}$'s determinant is nonzero.

In the equation above, the denominator is exactly $\mathbf{A}$'s determinant $\det(\mathbf{A})$; that is, when $ad-bc \neq 0$, $\mathbf{A}$ has an inverse matrix.

's denominator is $\mathbf{A}$'s determinant, so what is the numerator?

This numerator is called $\mathbf{A}$'s adjugate matrix (also known as the adjoint of a matrix), denoted $\text{adj}(\mathbf{A})$.

Thus, if the matrix $\mathbf{A}$ is invertible, $\mathbf{A}$'s inverse matrix is

$$\mathbf{A}^{-1} = \frac{\text{adj}(\mathbf{A})}{\det(\mathbf{A})}$$

That is, in order to compute the inverse matrix, we need to compute its determinant and its adjugate matrix.

The Adjugate Matrix

The adjugate matrix is defined as:

$$\text{adj}(\mathbf{A}) = \mathbf{C}^T$$

where $\mathbf{C}$ is called the cofactor matrix. Clearly, the adjugate matrix is the transpose of the cofactor matrix.

The Cofactor Matrix

The cofactor matrix is the matrix formed by the cofactor corresponding to each element of the matrix.

Each cofactor is obtained by deleting the row and column containing that element, taking the determinant of the resulting minor matrix (the minor), and then applying a sign adjustment, that is,

$$C_{i,j} = (-1)^{i+j} M_{i,j}$$

The minor $M_{i,j}$ is the determinant of the minor matrix that remains after deleting row i and column j from the matrix $\mathbf{A}$.

$$M_{i,j} = \det(\mathbf{A}_{i,j})$$

The minor matrix $\mathbf{A}_{i,j}$ is the matrix formed by the remaining elements of the matrix $\mathbf{A}$ after deleting row i and column j .

Let's look at the entire process of solving for the inverse matrix in reverse:

a) When expanding the determinant, extract the minor matrices (square matrices) needed for any row or column.

b) From the minor matrices (square matrices), compute the minors (scalars) for all elements of the original matrix;

- c) Compute the cofactor (scalar) for each element, that is, the sign-adjusted minor (scalar);
- d) Using Laplace expansion, compute the determinant.
- e) Using step c), construct the cofactor matrix.
- f) Transpose the cofactor matrix to obtain the adjugate matrix.
- g) Finally, multiply the reciprocal of the determinant by the adjugate matrix to obtain the inverse matrix.

 Note that when computing the determinant, you only need to expand along a specific row or column to extract minor matrices; but when constructing the cofactor matrix, you need to extract the minor matrix for every element.

By this point, many readers are probably feeling a bit dizzy.

Next, let's use the example from the last section of the previous chapter to fully compute the inverse of a 3×3 matrix.

A Complete Example

Given the following 3×3 square matrix $\mathbf{A}$, let's compute its inverse matrix step by step following the process above

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 0 & 1 \\ 1 & 2 & 1 \end{bmatrix}$$

You'll notice that most of the computations below come from the determinant-computation process in the last section of the previous chapter.

Extracting the Minor Matrices

Extract the minor matrix corresponding to each element $\mathbf{A}_{ij}$.

The minor matrix $\mathbf{A}_{1,1}$ (row 1, column 1 removed) is

$$\mathbf{A}_{1,1} = \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix}$$

The minor matrix $\mathbf{A}_{1,2}$ (row 1, column 2 removed):

$$\mathbf{A}_{1,2} = \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix}$$

The minor matrix $\mathbf{A}_{1,3}$ (row 1, column 3 removed):

$$\mathbf{A}_{1,3} = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix}$$

The minor matrix $\mathbf{A}_{2,1}$ (row 2, column 1 removed):

$$\mathbf{A}_{2,1} = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}$$

The minor matrix $\mathbf{A}_{2,2}$ (row 2, column 2 removed):

$$\mathbf{A}_{2,2} = \begin{bmatrix} 1 & 3 \\ 1 & 1 \end{bmatrix}$$

The minor matrix $\mathbf{A}_{2,3}$ (row 2, column 3 removed):

$$\mathbf{A}_{2,3} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}$$

The minor matrix $\mathbf{A}_{3,1}$ (row 3, column 1 removed):

$$\mathbf{A}_{3,1} = \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix}$$

The minor matrix $\mathbf{A}_{3,2}$ (row 3, column 2 removed):

$$\mathbf{A}_{3,2} = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$$

The minor matrix $\mathbf{A}_{3,3}$ (row 3, column 3 removed):

$$\mathbf{A}_{3,3} = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$$

Computing the Minors

Compute the determinant of each minor matrix $\mathbf{A}_{ij}$ that is, the minor M_{ij} .

The minor $M_{1,1}$:

$$M_{1,1} = \begin{vmatrix} 0 & 1 \\ 2 & 1 \end{vmatrix} = (0 \times 1) - (1 \times 2) = 0 - 2 = -2$$

The minor $M_{1,2}$:

$$M_{1,2} = \begin{vmatrix} 3 & 1 \\ 1 & 1 \end{vmatrix} = (3 \times 1) - (1 \times 1) = 3 - 1 = 2$$

The minor $M_{1,3}$:

$$M_{1,3} = \begin{vmatrix} 3 & 0 \\ 1 & 2 \end{vmatrix} = (3 \times 2) - (0 \times 1) = 6 - 0 = 6$$

The minor $M_{2,1}$:

$$M_{2,1} = \begin{vmatrix} 2 & 3 \\ 2 & 1 \end{vmatrix} = (2 \times 1) - (3 \times 2) = 2 - 6 = -4$$

The minor $M_{2,2}$:

$$M_{2,2} = \begin{vmatrix} 1 & 3 \\ 1 & 1 \end{vmatrix} = (1 \times 1) - (3 \times 1) = 1 - 3 = -2$$

The minor $M_{2,3}$:

$$M_{2,3} = \begin{vmatrix} 1 & 2 \\ 1 & 2 \end{vmatrix} = (1 \times 2) - (2 \times 1) = 2 - 2 = 0$$

The minor $M_{3,1}$:

$$M_{3,1} = \begin{vmatrix} 2 & 3 \\ 0 & 1 \end{vmatrix} = (2 \times 1) - (3 \times 0) = 2 - 0 = 2$$

The minor $M_{3,2}$:

$$M_{3,2} = \begin{vmatrix} 1 & 3 \\ 3 & 1 \end{vmatrix} = (1 \times 1) - (3 \times 3) = 1 - 9 = -8$$

The minor $M_{3,3}$:

$$M_{3,3} = \begin{vmatrix} 1 & 2 \\ 3 & 0 \end{vmatrix} = (1 \times 0) - (2 \times 3) = 0 - 6 = -6$$

Computing the Cofactors

A cofactor is simply a minor with a sign attached, following the rule $C_{i,j} = (-1)^{i+j} M_{i,j}$.

The cofactor $C_{1,1}$:

$$C_{1,1} = (-1)^{1+1} M_{1,1} = (1) \times (-2) = -2$$

The cofactor $C_{1,2}$:

$$C_{1,2} = (-1)^{1+2} M_{1,2} = (-1) \times (2) = -2$$

The cofactor $C_{1,3}$:

$$C_{1,3} = (-1)^{1+3} M_{1,3} = (1) \times (6) = 6$$

The cofactor $C_{2,1}$:

$$C_{2,1} = (-1)^{2+1} M_{2,1} = (-1) \times (-4) = 4$$

The cofactor $C_{2,2}$:

$$C_{2,2} = (-1)^{2+2} M_{2,2} = (1) \times (-2) = -2$$

The cofactor $C_{2,3}$:

$$C_{2,3} = (-1)^{2+3} M_{2,3} = (-1) \times (0) = 0$$

The cofactor $C_{3,1}$:

$$C_{3,1} = (-1)^{3+1} M_{3,1} = (1) \times 2 = 2$$

The cofactor $C_{3,2}$:

$$C_{3,2} = (-1)^{3+2} M_{3,2} = (-1) \times (-8) = 8$$

The cofactor $C_{3,3}$:

$$C_{3,3} = (-1)^{3+3} M_{3,3} = (1) \times (-6) = -6$$

Computing the Determinant

Expand along row 1 only to compute the determinant

$$\det(A) = a_{1,1}C_{1,1} + a_{1,2}C_{1,2} + a_{1,3}C_{1,3}$$

Substituting the values:

$$\det(A) = 1 \times (-2) + 2 \times (-2) + 3 \times 6 = -2 - 4 + 18 = 12$$

Constructing the Cofactor Matrix

Based on the results of through , the final cofactor matrix is $\mathbf{C}$

$$\mathbf{C} = \begin{bmatrix} C_{1,1} & C_{1,2} & C_{1,3} \\ C_{2,1} & C_{2,2} & C_{2,3} \\ C_{3,1} & C_{3,2} & C_{3,3} \end{bmatrix} = \begin{bmatrix} -2 & -2 & 6 \\ 4 & -2 & 0 \\ 2 & 8 & -6 \end{bmatrix}$$

Transposing to Obtain the Adjugate Matrix

Based on the definition of the adjugate matrix

$$\text{adj}(\mathbf{A}) = \mathbf{C}^T$$

Substituting in the specific values, we get $\text{adj}(\mathbf{A})$

$$\text{adj}(\mathbf{A}) = \begin{bmatrix} -2 & 4 & 2 \\ -2 & -2 & 8 \\ 6 & 0 & -6 \end{bmatrix}$$

Computing the Inverse Matrix

Based on

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \text{adj}(\mathbf{A})$$

Substituting in the specific values

$$\mathbf{A}^{-1} = \frac{1}{12} \begin{bmatrix} -2 & 4 & 2 \\ -2 & -2 & 8 \\ 6 & 0 & -6 \end{bmatrix}$$



LA_05_03_01.ipynb helps you master extracting minor matrices, computing minors and cofactors, solving the determinant, constructing the cofactor matrix, finding the adjugate matrix, and computing the inverse matrix. It also shows how to translate mathematical formulas into code. That said, from the standpoint of computational efficiency, stability, and numerical precision, it's still recommended to prioritize the linear algebra functions provided by NumPy for these computations.



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Use the method introduced in this section to compute the inverse of the following 2×2 matrix.

▶
$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Q2. Use the method introduced in this section to solve by hand for the inverse of the following 3×3 matrix, and verify the result using NumPy.

▶
$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

Q3. Please learn to use `time.time()` on your own to compare the computation time of `numpy.linalg.inv()` versus a custom function for solving the matrix inverse.

Q4. Please learn on your own how to solve a matrix inverse by hand using elementary row operations.